

ECON 351 - Intertemporal Consumption Pro

Screenshots from homework, sample exam, and mock midterm

Sources checked:

- Homework 1 Ch.4 Ex 5: 5.1-5.6 intertemporal consumption (Shohei contract)
- Sample Questions Ch.4: Q41-Q42 (save vs borrow, two interest rates)
- Mock Midterm 1: Q24-Q25 (budget constraint + optimum)
- Homework 2 Ch.9: No intertemporal questions
- Sample Questions Ch.9: No intertemporal questions
- Practice Tests 1-3: 2 intertemporal questions each (Q023-Q027 bank IDs)

5 problem sets in this PDF (all intertemporal problems found).

Intertemporal Consumption - Formula Sheet

Present-value budget: $c_1 + c_2/(1+r) = I_1 + I_2/(1+r)$

Future-value budget: $(1+r)c_1 + c_2 = (1+r)I_1 + I_2$

With unit prices $p_1=p_2=1$: $q_1 + q_2/(1+r) = I_1 + I_2/(1+r)$

Optimum (two goods): $MU_1/MU_2 = p_1/(p_2/(1+r)) = (1+r)$ when $p_1=p_2=1$

Cobb-Douglas $u = q_1 \cdot q_2$: at optimum $q_2 = (1+r)q_1$

Cobb-Douglas $u = c_1^a c_2^b$: $c_2/c_1 = (b/a)(1+r)$ at optimum

PV of income stream: $PV = I_1 + I_2/(1+r)$

Saving: spend less than I_1 in period 1 (lend). Borrowing: spend more than I_1 .

Problem 1: IT-1

Homework 1 Ch.4 - Exercise 5 (5.1 through 5.6)

Shohei contract problem - PV, MRS, optimum c_1/c_2 , contract comparison.

Exercise 5 - Intertemporal Consumption

In class, we saw how the baseball player Shohei was offered a peculiar contract. Here, we'll use this example to solve an intertemporal consumption problem.

5.1: Assume the Angels' offer him two contracts: the first pays him 10 million today and 90 million in the future, while the second pays him 42 million today and 54 million in the future. If the interest rate is $r = 0.20$, which contract is cheaper from the Angels' perspective?

5.2: Assume that Shohei, a very patient man, has the following utility function $U(c_1, c_2) = 2c_1^{0.4}c_2^{0.6}$ where c_1 is money for consumption today and c_2 is money for consumption in the future. In plain English, what does it imply that the exponent of c_2 is higher than the exponent of c_1 ? Calculate the ratio of the marginal utilities with MU_{c_1} in the numerator.

5.3: Now let's equalize the MRS to the price ratio, assuming unit prices. That is, $p_1 = p_2 = 1$. Remember to discount p_2 using the interest rates since c_2 is consumption in the future!

5.4: Assume the Angels offer Shohei the first contract. Find his optimum consumption c_1^* and c_2^* under that contract. Once you get them both, plug them into Shohei's utility function to find out how much utility he obtains from that contract.

5.5: Assume instead the Angels offer Shohei the second contract. Find his optimum consumption c_1^* and c_2^* under that contract. Once you get them both, plug them into Shohei's utility function to find out how much utility he obtains from this contract.

5.6 EXTRA: Which contract would Shohei prefer? If, for some reason, the Angels were low on funds today and were also unable to borrow, what type of contract could they offer Shohei such that he accepts it?

Problem 2: IT-2

Sample Questions Ch.4 - Question 41 | Practice Tests (Q025)

$I_1=10$, $I_2=12$, $r=0.5$, $u=q_1 \cdot q_2$. Find optimum and whether consumer saves.

Question 41:

Consider the following intertemporal consumption problem with one good and two periods. The quantity of the good consumed in period 1 and period 2 are q_1 and q_2 . The price of each unit of the good is \$1 in both periods. The consumer's income is $I_1 = 10$ in the first period and $I_2 = 12$ in the second period. The consumer can borrow or save money at the interest rate of 50%, that is, $r=0.5$. The consumer's utility function is $u(q_1, q_2) = q_1 q_2$.

What is the optimal choice of q_1 and q_2 ? Is this consumer saving or borrowing? How much?

Problem 3: IT-3

Sample Questions Ch.4 - Question 42

Same setup as Q41 but $r=0.05$. Consumer borrows instead of saves.

Question 42:

Consider the following intertemporal consumption problem with one good and two periods. The quantity of the good consumed in period 1 and period 2 are q_1 and q_2 . The price of each unit of the good is \$1 in both periods. The consumer's income is $I_1 = 10$ in the first period and $I_2 = 12$ in the second period. The consumer can borrow or save money at the interest rate of 5%, that is, $r=0.05$. The consumer's utility function is $u(q_1, q_2) = q_1 q_2$.

Problem 4: IT-4

Mock Midterm 1 - Question 24 | Practice Tests (Q026)

$I_1=60$, $I_2=125$, $r=0.25$, $u=q_1 \cdot q_2$. Identify the intertemporal budget constraint.

Question 24:

[Show your work]

Consider the following intertemporal consumption problem with one good and two periods. The quantity of the good consumed in period 1 and period 2 are q_1 and q_2 . The price of each unit of the good is \$1 in both periods. The consumer's income is $I_1 = 60$ in the first period and $I_2 = 125$ in the second period. The consumer can borrow or save money at the interest rate of 25%, that is, $r=0.25$. The consumer's utility function is $u(q_1, q_2) = q_1 q_2$.

The consumer's inter-temporal budget constraint is

- (A) $q_1 + 0.8q_2 = 160$
- (B) $q_1 + q_2 = 185$
- (C) $q_1 + 1.25q_2 = 216.25$
- (D) $q_1 + 0.75q_2 = 160$
- (E) $q_1 + 0.25q_2 = 200$

Problem 5: IT-5

Mock Midterm 1 - Question 25 | Practice Tests (Q027)

Using Q24 setup, find optimal q_1 in period 1.

Question 24:

[Show your work]

Consider the following intertemporal consumption problem with one good and two periods. The quantity of the good consumed in period 1 and period 2 are q_1 and q_2 . The price of each unit of the good is \$1 in both periods. The consumer's income is $I_1 = 60$ in the first period and $I_2 = 125$ in the second period. The consumer can borrow or save money at the interest rate of 25%, that is, $r=0.25$. The consumer's utility function is $u(q_1, q_2) = q_1 q_2$.

The consumer's inter-temporal budget constraint is

- (A) $q_1 + 0.8q_2 = 160$
- (B) $q_1 + q_2 = 185$
- (C) $q_1 + 1.25q_2 = 216.25$
- (D) $q_1 + 0.75q_2 = 160$
- (E) $q_1 + 0.25q_2 = 200$

Question 25:

[Show your work]

Consider all the information from Question 24. In the first period, the consumer will optimally purchase

- (A) $q_1 = 0$
- (B) $q_1 = 100$
- (C) $q_1 = 60$
- (D) $q_1 = 80$
- (E) $q_1 = 120$

Answer Key

IT-1 (5.1):

$PV1 = 10 + 90/1.2 = 85M$; $PV2 = 42 + 54/1.2 = 87M$. Contract 1 is cheaper.

IT-1 (5.2):

Higher exponent on c_2 means Shohei values future consumption more (patient). $MU_{c1}/MU_{c2} = (2/3)(c_2/c_1)$.

IT-1 (5.3):

$MRS = (1+r) = 1.2$, so $c_2 = 1.8 c_1$.

IT-1 (5.4):

Budget: $c_1 + c_2/1.2 = 85$. With $c_2=1.8c_1$: $c_1^* = 34$, $c_2^* = 61.2$.

IT-1 (5.5):

Budget: $c_1 + c_2/1.2 = 87$. $c_1^* = 34.8$, $c_2^* = 62.64$.

IT-1 (5.6):

Compare utility from each contract; Angels could offer a front-loaded contract if cash-constrained.

IT-2 (Q41):

$q_1^* = 9$, $q_2^* = 13.5$. Consumer saves 1 (spends $9 < I_1=10$).

IT-3 (Q42):

$q_1^* = 75/7$, $q_2^* = 45/4$. Consumer borrows $5/7$ (spends more than I_1 in period 1).

IT-4 (Mock Q24):

(A) $q_1 + 0.8q_2 = 160$. PV budget with $1/(1+r) = 0.8$.

IT-5 (Mock Q25):

(D) $q_1 = 80$. Cobb-Douglas $u=q_1q_2$ gives $q_2=(1+r)q_1$; budget yields $q_1=80$.